
Instructions for the student:

1. Attempt any five questions.
 2. Illustrate your answer with neat sketches, diagrams etc. wherever necessary.
 3. Necessary data is given in the respective questions. If such data is not given, it means that the knowledge of that component is a part of examination.
 4. If some part or parameter is noticed to be missing, you may appropriately assume and state it clearly in the answer-book.
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- Q.1. A] Define AM. Derive the expression of AM wave. How the total power and total current in AM wave is estimated. 05
- B] The antenna current of an AM transmitter with 40% MI is 11A. It increases to 12A as a result of simultaneous modulation by another sine wave. What is the MI due to second wave? 05
- Q.2. A] Draw and explain ISB in detail. 05
- B] What is SSB? Explain, with suitable derivation, how suppression of carrier takes place in SSB 05
- Q.3. A] Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of (a) 100% and (b) 50%. 05
- B] Draw and explain the phase-shift method for SSB. 05
- Q.4. A] In an FM system, when audio frequency is 500Hz and the AF voltage is 2.4V, the deviation is 4.8KHz. If the AF voltage is now increased to 7.2V, what is the new deviation? If the AF voltage is now increased to 10V while the AF is dropped to 200Hz, what is the deviation? Calculate the modulation index in each case. 05
- B] Explain the telemetry system in detail. 05

- Q.5. A] The equation of an angle-modulated voltage is $v = 10\sin(10^8t + 3\sin10^4t)$. 05
What form of angle modulation is this? Calculate the carrier and modulating frequencies, the modulation index and deviation, and the power dissipated in 100Ω resistor.
- B] Explain the sky wave propagation in detail. Define different criteria's which 05
decides the quality of sky wave propagation.
- Q.6. A] Calculate the image rejection of a receiver having an RF amplifier and an 05
IF of 450KHz; if the Q_s of the relevant coils are 65 at an incoming frequency of (a) 1200KHz and (b) 20MHz.
- B] Define and classify a loudspeaker. Explain a moving coil loudspeaker in 05
detail along with its technical specifications.
- Q.7. A] List different techniques of FM generation. Explain any one technique in detail. 05
- B] What is PCM? Discuss how the step size affects the quality of PCM. Compare 05
PCM and DM on any four functional criteria's. <https://www.batuonline.com>
- Q.8. A] Explain the role of pre-emphasis and de-emphasis in FM communication. 05
- B] Calculate the MUF for a communication system operating with a critical 05
frequency of 1GHz and angle of incidence 40° .
